An Image Is Worth 16 × 16 Words: Transformers for Image Recognition at Scale

1 Motivations

1. While the Transformer architecture has become the de-facto standard for natural language processing tasks, its application to computer vision remains limited.

2. Prior work applied attention in conjunction with convolutional networks; this paper shows that a pure Transformer, applied directly to sequences of image patches, can perform very well on image classification tasks.

2 Solution

2.1 Model Architecture

The main concept is to follow the original Transformer as closely as possible. The detailed steps of the Vision Transformer (ViT) are:

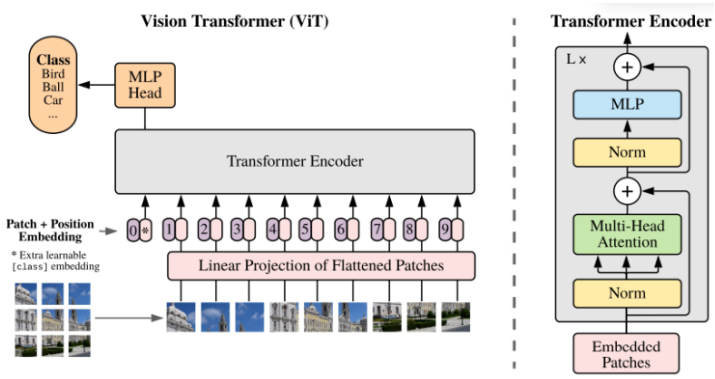


Figure 1: Vision Transformer Architecture

1. Reshape the image $x \in \mathbb{R}^{H \times W \times C}$ into a sequence of flattened 2D patches $x_p \in \mathbb{R}^{N \times (P^2 C)}$, where $N = \frac{H W}{P^2}$.

2. Apply a trainable linear projection E and add a learnable positional encoding E_{pos}:

$$z_0 = \left[\mathbf{x}_{\text{class}};\ x_p^1 E;\ \ldots;\ x_p^N E\right] + E_{\text{pos}}, \quad E \in \mathbb{R}^{(P^2C)\times D},\ E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D}.$$

3. Repeat L times:

$$z'_l = \text{MSA}\big(\text{LN}(z_{l-1})\big) + z_{l-1}, \quad l = 1, \ldots, L.$$

4. Then:

$$z_l = \text{MLP}\big(\text{LN}(z'_l)\big) + z'_l, \quad l = 1, \ldots, L.$$

5. After L encoder blocks and a final LayerNorm, output the classification token:

$$y = \text{LN}\big(z_L^0\big).$$

2.2 Model Variants

Table 1: Vision Transformer Variants

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

(Here, "ViT-L/16" means the Large variant with 16×16 patch size.)

2.3 Training Parameters

All models use batch size 4096, Adam optimizer, and a 10k-step warmup.

3 Evaluation Method

3.1 Training Datasets

- ILSVRC-2012 ImageNet (1k classes, 1.3M images)
- ImageNet-21k (21k classes, 14M images)
- JFT (18k classes, 303M images)

Table 2: Training Configurations

Models	Dataset	Epochs	BaseLR	LR decay	Weight decay	Dropout
ViT-B/16,32	JFT-300M	7	8×10^{-4}	linear	0.1	0.0
ViT-L/32	JFT-300M	7	6×10^{-4}	linear	0.1	0.0
ViT-L/16	JFT-300M	7/14	4×10^{-4}	linear	0.1	0.0
ViT-H/14	JFT-300M	14	3×10^{-4}	linear	0.1	0.0
R50×1,2	JFT-300M	7	1×10^{-3}	linear	0.1	0.0
R101×1	JFT-300M	7	8×10^{-4}	linear	0.1	0.0
R152×1,2	JFT-300M	7	6×10^{-4}	linear	0.1	0.0
R50+ViT-B/16,32	JFT-300M	7	8×10^{-4}	linear	0.1	0.0
R50+ViT-L/32	JFT-300M	7	2×10^{-4}	linear	0.1	0.0
R50+ViT-L/16	JFT-300M	7/14	4×10^{-4}	linear	0.1	0.0
ViT-B/16,32	ImageNet-21k	90	1×10^{-3}	linear	0.03	0.1
ViT-L/16,32	ImageNet-21k	30/90	1×10^{-3}	linear	0.03	0.1
ViT-*	ImageNet	300	3×10^{-3}	cosine	0.3	0.1

3.2 Fine-tuning Benchmarks

- ImageNet (validation & ReaL labels)
- CIFAR-10/100
- Oxford-IIIT Pets & Flowers-102
- VTAB-19 suite

4 My Analysis

This work applies a pure Transformer to vision tasks, showing that with large-scale pre-training, ViT outperforms CNNs. ViT excels in global context modeling but is data-hungry.

1. **Global vs. Local Receptive Field:**
   A Transformer can attend to the entire image at once, whereas a CNN only "sees" a small local window in early layers. This global context is the biggest advantage of the Transformer.

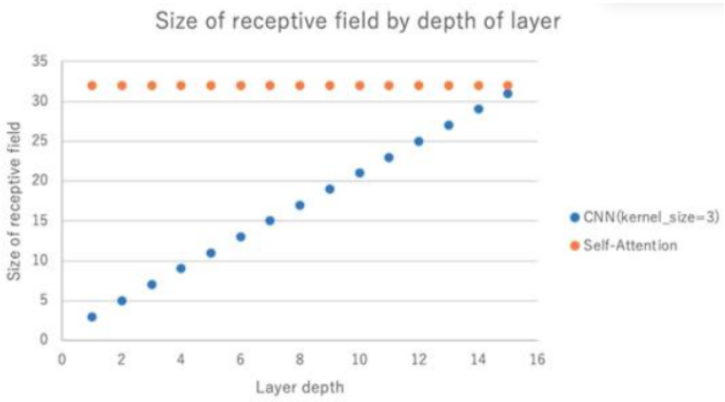


Figure 2: Size of receptive field by depth of layer: CNN vs. Self-Attention

2. **Data Dependence:**
   Performance of Transformers is heavily tied to training dataset size. With larger datasets, Transformers outperform CNNs by a wider margin.

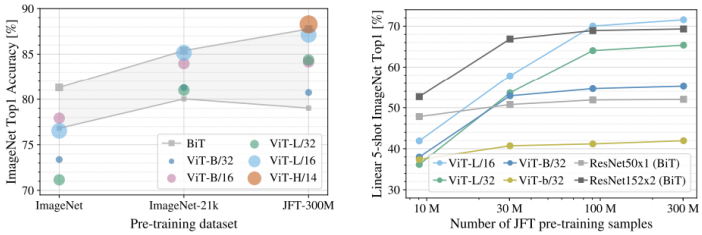


Figure 3: Transformer vs. CNN accuracy as dataset size increases

3. **Hybrid Models:**
   Hybrid architectures (e.g., CNN–Transformer) improve on pure Transformers at smaller model scales, but the performance gap closes as model size increases.

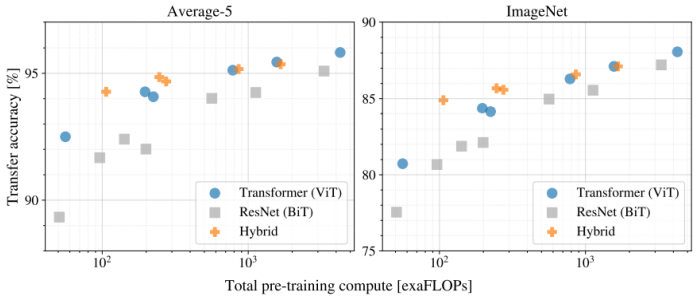


Figure 4: Hybrid vs. Pure Transformer performance across model sizes

5 Future Works

1. **Data-Efficient ViTs**: Incorporate locality biases to reduce data requirements.
2. **Efficient Attention Mechanisms**: Develop sparse or linear-complexity attention.
3. **Hierarchical & Multi-Scale Architectures**: Build multi-scale ViT backbones.
4. **Self-Supervised & Unsupervised Pre-Training**: Leverage unlabeled data.
5. **Dense Prediction Applications**: Adapt ViT for detection and segmentation.

6 My Questions

1. How can ViT be best adapted for detection, instance segmentation, and semantic segmentation?
2. Which sparse attention mechanisms work for very large images?
3. What is the minimal pre-training dataset size needed for ViT to match CNNs?